Charitha Ramireddy

Full Stack Developer | MERN | Real-time Voice & AI Systems

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SUMMARY

Full Stack Developer with **5+ years** of experience on the **MERN stack (MongoDB, Express.js, React.js, Node.js)**, building B2C and B2B cloud telephony products (**Doosra** and **Alohaa**) end to end. Builds complex, performant **React.js UIs and dashboards** on well-designed backends: Node.js microservices, RabbitMQ pipelines, Redis caching and MongoDB schemas. Specializes in **real-time voice systems**, including the WebSocket voice streaming infrastructure that powers LLM voice bots and AI call analysis. Started at **EY (Big Four)**; **IIIT Sri City** graduate (Institute of National Importance) with a **9.06 CGPA**.

SKILLS

Languages: JavaScript (ES6+), Python
Frontend: React.js, Redux, React Hooks, Material UI, Socket.IO, WebRTC
Backend: Node.js, Express.js, REST APIs, Microservices, JWT Authentication, OAuth 2.0, Webhooks
Architecture: Event-Driven Architecture, Message Queues, Rate Limiting, Retry Handling
Database & Messaging: MongoDB (Mongoose, Aggregation Pipelines, Indexing, Schema Design), Redis, RabbitMQ, Cron Jobs
AI & Voice: OpenAI (GPT-4o, Whisper), Groq, ElevenLabs, Speech-to-Text, Text-to-Speech, Asterisk, IVR, Voice Bots, Voice Streaming
Integrations: Zoho CRM, HubSpot, Zendesk, LeadSquared, Slack, Razorpay
Cloud & DevOps: AWS (S3), Docker, PM2, Bitbucket CI/CD, Unleash (Feature Flags), Sentry, Git
Tools: Jira, Postman, Claude Code

EXPERIENCE

Full Stack Developer | Matr Innovations, Hyderabad — Jul 2024 – Present

Core developer on **Alohaa**, a B2B cloud telephony SaaS platform used by **50+ businesses** and **1,000+ agents**, handling **1.1M+ calls per day** across voice APIs, broadcast campaigns and AI voice streaming.

- Designed and built **Node.js microservices** for a voice and SMS broadcast platform (4 services) sending **1M+ messages per day** across **1,000+ campaigns per month**, covering campaign scheduling, bulk lead upload, credit-based billing and rate-limited dispatch through **RabbitMQ**.
- Developed **REST APIs** for a programmable voice API handling **500K calls per day** (750K peak), with JWT authentication, webhook callbacks and IVR analytics; added **Redis caching** that cut API response time from **~200 ms to ~5 ms**.
- Built the **WebSocket voice streaming gateway** connecting Asterisk calls to real-time AI voice bots (streaming STT, LLM and TTS), now handling **100K+ calls per day** (200K peak); helped integrate **Groq** for LLM inference and **ElevenLabs** for TTS.
- Migrated call flows onto the **event-driven** post-call pipeline (RabbitMQ topic exchange, **~5M messages per day**), building retry consumers that raised **webhook delivery success from 95% to 99%** without ever blocking live calls.
- Built **React.js dashboards** for AI call quality analysis (Whisper transcripts, GPT-4o summaries, sentiment, top and bottom performing agents) and campaign analytics reports, powered by **MongoDB aggregation pipelines**.
- Integrated **Zoho CRM, HubSpot, Zendesk and LeadSquared** to auto-sync call logs, recordings and contacts for **10+ clients**, saving agents hours of manual logging per day.
- Worked directly with clients to gather requirements and shipped features **end to end** (React UI, Node.js/Express APIs, MongoDB schemas); deployed with **Docker** and Bitbucket CI/CD, with Sentry error tracking.

Full Stack Developer | Ten20 Infomedia, Hyderabad — Apr 2022 – Jun 2024

Developer on **Doosra**, a B2C cloud telephony app with **10K+ users**, **20K+ calls** and **40K+ messages per day**, building features across the React.js frontend and Node.js/Express backend.

- Drove multiple **cost optimizations** by analyzing paid third-party services and replacing them with in-house solutions, **saving $1,500 per month** ($18K per year); e.g., replaced the paid Google classifier used for SMS classification.
- Built **React.js dashboards** and metrics for call, contact and message analytics end to end, with Node.js APIs and performance-optimized **MongoDB aggregation pipelines**, supporting **weekly and monthly analysis** for internal teams and end users.
- Released features behind **Unleash feature flags**; automated billing, reports and data cleanup with **cron jobs**.

Intern & Trainee, Financial Accounting Advisory Services (FAAS) | Ernst & Young (EY), Big Four — May 2018 – Nov 2018

- Built the **financial reporting module** in **DIGIFAAS**, EY's in-house accounting platform, automating the generation of **balance sheets** and **Profit & Loss (P&L) statements**.
- Worked inside EY's FAAS practice, learning **financial reporting**, accounting workflows and enterprise-grade delivery standards early in my career.

EDUCATION

B.Tech, Electronics & Communication Engineering | Indian Institute of Information Technology (IIIT) Sri City — 2014 – 2018

- Graduated with a **CGPA of 9.06 / 10** from an **Institute of National Importance**.
- Selected as **Teaching Assistant** for Mathematics, Basic Electronic Circuits and VLSI while still an undergraduate.